Computer Ethics, Ethical Theory, Professional Responsibility and Privacy Online

Part I — Introduction to Computer Ethics

Computer ethics examines the moral dimensions of computing — the decisions made by those who design, deploy, and use technology, and the consequences those decisions carry for individuals and society.

James Moor's foundational insight (1985) is that computers are logically malleable — configurable for virtually any task — and this creates policy vacuums: situations where existing laws and social norms are absent or inadequate. When email was invented, there was no settled norm about whether an employer could read an employee's messages. When social media emerged, there were no agreed rules about defamation or harassment in a networked public sphere. Computer ethics fills these vacuums.

Why it matters for engineers: Technology encodes values. A recommendation algorithm that optimizes for engagement is implicitly judging that engagement matters more than accuracy. A facial recognition system trained on biased data makes implicit judgments about whose face deserves correct identification. These are ethical choices — whether or not the engineers recognized them as such.

Ethics vs. Law: Legal compliance is the floor of ethical obligation, not the ceiling. Many data collection practices that are technically lawful would be regarded as exploitative if users understood them. "We're compliant" is never a complete answer to an ethical challenge.

Part II — Ethical Theories

No single theory provides a complete answer to every moral problem. The productive approach is to use them together.

Consequentialism

The right action is the one that produces the greatest good for the greatest number. Outcomes are what matter.

Example: A security team decides not to patch a minor vulnerability because the cost of patching (downtime, engineering hours) outweighs the low probability of exploitation. That is consequentialist reasoning.

The problem: It can justify harming individuals for aggregate benefit. A system that harvests patient data without consent to train a better cancer diagnostic model produces good outcomes — but most people would say something is still wrong with it. Aggregate improvement does not dissolve individual rights.

Deontology (Kant)

The right action is one that conforms to moral duties, regardless of consequences. Kant's key rule: treat persons always as ends in themselves, never merely as means.

Example: Dark patterns — interface designs that trick users into sharing more data or subscribing to services they didn't intend to — are deontologically wrong because they manipulate users rather than respect their rational agency, even if no measurable harm results.

The strength: It resists trading away individual rights for aggregate gain. The limitation: Rigid rules can seem counterproductive at the margins, which is why most ethicists use deontological constraints as boundaries on consequentialist reasoning rather than absolute rules.

Virtue Ethics

Shifts the question from what should I do? to what kind of person should I be? It emphasizes character and practical wisdom (phronesis) — the judgment to act well in complex, rule-underdetermined situations.

Example: How much effort should an engineer invest in security testing beyond the minimum required? Rules and outcomes both underdetermine the answer. A person of professional integrity invests the effort that genuine care for users warrants — not the minimum they can get away with.

Relevant professional virtues: intellectual honesty, professional courage (raising concerns when uncomfortable), fairness, and care for those affected by one's work.

Contractualism (Rawls)

Just principles are those that rational people would choose if they did not know their position in society — behind a "veil of ignorance". You don't know if you'll be the data subject or the data collector, the citizen under surveillance or the surveillance operator.

Example: Would you consent to algorithmic credit scoring if you didn't know whether you'd be in the majority group that benefits or the minority group that is systematically disadvantaged? This thought experiment identifies unfairness that aggregate statistics can obscure.

Using Multiple Frameworks

When frameworks converge on a conclusion, confidence increases. When they diverge, the divergence reveals a genuine moral tension. Consider the tradeoff between contact tracing apps during a pandemic and individual privacy: consequentialism favors deployment, deontology demands consent and proportionality, virtue ethics asks whether authorities are acting in good faith, and contractualism asks whether the least powerful would agree to these terms. None of those questions is irrelevant.

Part III — Professional Ethics

What Makes a Profession

A profession is distinguished by specialized knowledge, significant autonomy in practice, a claim to serve the public interest, and self-regulatory mechanisms including codes of ethics. Computing acquired mass social influence faster than it developed the institutional apparatus of professional self-regulation — which is part of why explicit ethical reasoning matters so much in this field.

The ACM Code of Ethics (2018)

The ACM Code establishes obligations around honesty, avoiding harm, respecting privacy, and acting fairly. Its 2018 revision importantly extended professional responsibility to indirect and systemic harm — not just immediate physical safety. A system that works exactly as designed can still cause serious harm through discriminatory outcomes or privacy erosion.

Example: An engineer who builds a hiring algorithm that correctly predicts performance on average, but inadvertently screens out candidates from certain zip codes correlated with race, has caused harm — even if no individual decision was manually biased. The 2018 code explicitly covers this.

The Dual Obligation Problem

Computing professionals are employed by organizations whose obligations run to shareholders, yet their work directly affects users who have no voice in that organization. The ACM code resolves this by treating public welfare as prior to organizational interest — but this is easier to state than to apply.

Example: An engineer discovers that a product's default privacy settings expose more user data than most users would want, but the setting increases ad revenue. Changing it requires making the case internally against financial pressure. "Legal approved it" ends the legal analysis but not the ethical one.

Whistleblowing

When internal remediation has failed and harm is serious, disclosure to external parties may be ethically required. Both deontological duties of non-maleficence and consequentialist concern for harm point the same direction. Virtue ethics reinforces this as professional courage — the willingness to bear personal cost in service of moral obligation.

Example: Frances Haugen's disclosure of Facebook's internal research on the platform's harm to teenage girls is a contemporary case where an engineer/analyst concluded internal channels were exhausted and the harm was serious enough to justify going public.

Part IV — Privacy: A Contextual Approach

Why Privacy Matters

Privacy is a precondition for autonomy — the capacity to form and act on one's own values without undue surveillance or interference. The loss of control over personal information is a harm in itself, because the capacity for self-determination is damaged when informational self-governance is lost.

The Secrecy Model (and Why It Fails)

The common view equates privacy with secrecy: once information is disclosed to anyone, privacy claims are extinguished. This is the logic behind "nothing to hide, nothing to fear."

It fails on both counts. Empirically, people share information in contexts governed by clear norms about further use — sharing your symptoms with a doctor is not consent for that information to flow to your employer. Normatively, the model cannot explain why many obvious privacy violations are wrong: if a stranger systematically follows you in public — observing only what is visible — the secrecy model says no violation has occurred. Intuition disagrees.

Contextual Integrity (Nissenbaum, 2010)

Privacy is not about secrecy — it is about appropriate information flow, where "appropriate" is defined by the norms of the context in which information was originally shared.

Every social context operates under norms that specify what information is relevant, to whom it flows, and on what terms (transmission principles). A privacy violation occurs when information flows in ways that violate the norms of the context in which it was shared.

Example: A patient discloses a chronic condition to their physician. The transmission principle is medical confidentiality: the information may flow to other treating clinicians on a need-to-know basis, but not to the patient's employer, not to a pharmaceutical marketing database. If it flows to either of those, a privacy violation has occurred — not because the information was secret, but because its flow violated the norms of the medical context.

The aggregation problem. Contextual integrity explains why combining individually innocuous data is still a privacy violation. Your name, home address, employer, and daily commute route are each arguably public. Combined and sold to a data broker without your knowledge, the aggregate violates the norms of every context from which a piece came — none of which authorized this recombination.

Example: In 2006, AOL released "anonymized" search query logs. Researchers re-identified individual users by cross-referencing queries — a person who searched for their own name, local restaurants, and specific medical conditions was uniquely identifiable within days. No individual query was secret; the aggregation destroyed privacy.

Online Contexts Where Norms Are Violated

Third-party tracking. When you read about a health condition on WebMD, purchase something on Amazon, and read political commentary on a news site, you have shared information in three entirely separate contexts with incompatible norms. Ad networks that track you across all three and combine the data are routing information across contexts in ways that none of those contexts authorized. This is what contextual integrity identifies as wrong, regardless of whether it is technically disclosed in a terms-of-service agreement.

Context collapse on social media. Platforms flatten multiple audiences — close friends, professional contacts, family, strangers — into a single undifferentiated public. A post written for friends may reach an employer; a joke in one social context lands differently in another. Platforms that make this collapse the default, and granular control difficult, are making a design choice with ethical implications.

Workplace surveillance. Logging employee keystrokes, monitoring email content, or tracking physical location moves information collected in a work context into a disciplinary context with different norms. Disclosing the monitoring in an employment contract does not make the information flow contextually appropriate — disclosure of surveillance is not the same as the norms of the work context authorizing that surveillance.

Re-identification: Anonymization Is Not a Solution

Latanya Sweeney demonstrated in 2000 that 87% of the US population can be uniquely identified by ZIP code, birthdate, and sex alone — all routinely included in "anonymized" datasets. Mobile location traces can be re-identified with as few as four spatio-temporal data points. Netflix's anonymized viewing history dataset was de-anonymized by cross-referencing with public IMDb reviews.

The practical implication: claims that data has been anonymized should be treated with skepticism proportional to the richness of the dataset and the availability of external data for cross-referencing. The ethical question is not whether a dataset is technically anonymized, but whether individuals could be re-identified under realistic conditions.

Legal Frameworks and Their Limits

The GDPR (EU, 2018) is the most comprehensive modern framework: data minimization, purpose limitation, rights of access and erasure, and significant enforcement penalties. The US relies on a sectoral patchwork — HIPAA for health, FERPA for education, COPPA for children under 13 — leaving behavioral data, location data, and inferred data largely unprotected at the federal level.

The gap between legal permissibility and ethical acceptability is wide. Much of what contextual integrity identifies as privacy violations is technically lawful. This reinforces the opening point: legal compliance is the floor, not the ceiling.